

ASSIGNMENT COVER SHEET

PROGRAMME : MASTER OF BUSINESS ANALYTICS

SUBJECT CODE AND TITLE : BAA5123 AI AND DECISION MAKING

ASSIGNMENT TITLE : Enhancing Dairy Retail Operations through Artificial Intelligence and Advanced Analytics

LECTURER : Dr Mohammed Al-Obaydee ASSIGNMENT DUE DATE: 11 Dec 2025

STUDENT'S DECLARATION

- I hereby declare that this assignment is based on my own work except where acknowledgement of sources is made.
- I also declare that this work has not been previously submitted or concurrently submitted for any other courses in Sunway University/College or other institutions.
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[☒] I / We used the following A.I. tools to produce content in this submission:

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Grammarly	Correct grammar and spelling, improve sentence structure and recommend alternative vocabulary choices to enrich the language and tone of the essay.	N/A	Grammarly suggestions were used for all sections of the essay.
QuillBot	Paraphrasing and rewording of draft content for clarity and improved expression.	N/A	Several sentences in Introduction and Literature Review parts were paraphrased to improve fluency and coherence. The original analysis and conclusions are entirely the writer's own.
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	Summarising	Please summarize this journal article /paragraph	Some of the journal articles used in Literature Review parts were summarized by AI.
	Assisting in structuring points	Please suggest a structure for my essay with the relevant points	The structure for Introduction part was inspired by AI.

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Enhancing Dairy Retail Operations through Artificial Intelligence and Advanced Analytics

1. Introduction

The operational environment for dairy retailers is one of the most demanding in the fast-moving consumer goods (FMCG) industry. The highly perishable, short shelf life, and thermosensitive constraints of the dairy products, such as milk, cheese, butter, and yogurt have put enormous pressure on retailers, who must ensure adequate stock on their shelves while preventing spoilage and waste in order to protect their profits. Given these constraints, the accuracy of demand forecasting becomes essential as minor inaccuracies can lead to two negative outcomes, either understocking which results in lost sales and customer dissatisfaction, or overstocking, which leads directly to expired goods and financial loss.

In this study, we have used the Dairy Product Sales dataset from the Kaggle platform (Suraj, 2023). This dataset provides a realistic, multidimensional view of dairy retail operations. It includes farm-level production factors and detailed product attributes like shelf life. The dataset also captures real-time inventory conditions and actual sales transactions across locations and channels. Collectively, this data illuminates the core operational challenges in dairy retail, and the inherent difficulties have also been revealed in the inventory planning, shelf-life management, and setting competitive yet profitable prices.

These operational challenges in the retail industry are intensifying and are now recognized as critical business risks. Industry reports indicate the three strategic imperatives that have become industry necessities, which are reducing food waste, tighten supply-chain responsiveness, and fulfilling sustainability commitments to stakeholders. On the demand side, rapidly changing dietary trends, persistent inflation, and significantly increased consumer price sensitivity have made purchasing behaviour highly uncertain and unpredictable. The vulnerability of traditional supply-side operating models has been directly highlighted by this unpredictability. Rigid inventory management systems, replenishment models relying on manual decision-making, and static planning based on historical data are no longer effective in responding to real-time demand fluctuations. The result is a worsening supply-demand mismatch, manifested in missed opportunities, increased waste due to excess inventory, and customer churn due to stockouts or excessively high prices. Therefore, the fundamental solution to addressing the unpredictability of consumer behaviour lies in the digital and intelligent transformation of the supply chain.

In this context, the Artificial Intelligence (AI) plays a crucial role in providing data-driven decision support across multiple areas of retail operations. It enhances retailers' ability to forecast the demand more accurately, assessing product expiration dates more reliably, and analyzing the precise impact of pricing on profitability. By transforming raw historical data into insightful forward-looking metrics, AI helps businesses optimize inventory levels, plan replenishment cycles, and implement strategic pricing. This case study explores how AI-driven analytics can help retailers build a more efficient, sustainable, and profitable dairy supply chain.

2. Literature Review

The retail industry, especially the retail of perishable goods such as dairy products, is facing multiple challenges including fluctuating demand, inventory losses, pricing optimization, and supply chain efficiency. The traditional decision-making models become inadequate mainly due to the time-sensitivity of products, the products' sensitivity to seasonality and consumer behaviour. In recent years, the development of data analytics and AI technologies has provided new pathways to address these challenges. It is a vital tool used to enhance demand forecasting, inventory control, and dynamic pricing by converting complicated data into predictive insights. This strengthens supply-chain resilience and increases profitability in the dairy and FMCG industries. Therefore, this literature review examines recent research across four interrelated themes, which are demand forecasting, inventory and expiry management, pricing strategy, and the role of AI in retail decision support, highlighting both progress and unresolved challenges.

2.1 Perishable Goods Demand Forecasting in Retail

Demand forecasting remains a challenging task for perishable goods in the retail industry due to volatile consumer demand, seasonality, promotional activities, and perishability constraints (Fatorachian & Shokri, 2025). According to a 2024 study by Zhu, dynamic sales trends can be captured by time-series-based forecasting techniques in various retail environments and improve forecasting accuracy in unstable environments. Similarly, Ye et al. (2024) pointed out that Fourier time-varying grey models provide better forecasting for seasonal retail demand. These models take into account the patterns of change that are often ignored by traditional methods (Jauhar et al., 2024; Komijani and Sajadieh, 2024). Despite these advances, several challenges remain. These include the need for high-frequency data, accurate integration of external factors, and improved demand forecasting for individual perishable products instead

of aggregated categories. To simultaneously control dairy product spoilage costs and ensure timely replenishment, accurate demand forecasting is crucial. Dairy product sales data provides an excellent research scenario for this, allowing us to model and analyse at the individual product level, thereby driving operational planning from experience-driven to forecast-driven.

2.2 Perishable Inventory and Expiry Management

To reduce food waste and ensure supply chain efficiency, precise management of inventory and shelf life is critical. According to Birkmaier et al. (2024) and Komijani and Sajadieh (2024), an integrated planning approach reduces spoilage risk. This method combines production, inventory, and route decisions. Cold-chain predictive models can provide early warnings of potential spoilage. These warnings enable proactive interventions to prevent waste. In support of this, Fatorachian and Pawar (2025) demonstrated that by matching inventory with demand, the risks of overproduction and spoilage can be significantly reduced by AI-driven predictive models — especially ARIMA and multiple linear regression (MLR), thereby improving the resilience of the supply chain. In addition, Korie (2025) also highlighted the role of AI-driven monitoring. These predictive management systems optimize inventory turnover and reduce expiration-related losses. Nevertheless, significant limitations persist. Current models often lack universality, struggling to handle diverse products with different shelf lives. Another challenge is the technical complexity of integrating real-time supply chain data with predictive algorithms. This study addresses these gaps by focusing on the dairy sector. It utilizes AI for classification and prediction to optimize inventory and cut expiration-related losses.

2.3 Pricing Strategy and Revenue Optimization for FMCG

Pricing strategy and revenue optimization in FMCG retail are closely tied to demand patterns, product perishability, and competitive pressures. According to Korie (2025), integrating AI for dynamic pricing and demand forecasting creates a systematic solution, cutting perishable waste by as much as 30% and mitigating overstock issues. These results align with established research, which underscores that precise forecasting is fundamental to aligning production and procurement with actual market demand. Despite this, a critical literature gap remains. There is a lack of comprehensive decision-making frameworks designed specifically for perishable goods. This is particularly true for frameworks that account for stochastic, or variable, product lifetimes. This study is relevant for its integrated approach to a defined problem. It combines demand forecasting, inventory management, and pricing analytics to optimize revenue in the

perishable dairy sector, which faces distinct challenges of storage variability and high wastage (Komijani & Sajadieh, 2024).

2.4 Role of AI and Data-Driven Decision Support in Retail

The adoption of AI and data-driven systems in retail and supply chain management has accelerated significantly. These technologies now offer transformative benefits. They enhance capabilities in forecasting, inventory management, and pricing strategy. Studies point to two critical technological advances for supply chain optimization. Birkmaier et al. (2024) demonstrated that better replenishment planning stems from integrating external data with explainable AI. Complementing this, and Maheshwari et al. (2023), Falayi et al. (2024), and Duttyal (2025) emphasized the role of digital twins in providing real-time monitoring and control, particularly for perishable goods. Additionally, Korie (2025) also highlighted the role of AI and predictive analytics in reducing global food waste. This aligns the core operational efficiency with key sustainability goals. There are still major obstacles to overcome, like data silos and making sure solutions are tailored to specific industry contexts (Falayi et al., 2024; Ye et al., 2024). This study leverages AI on the Dairy Product Sales dataset to tackle the sector's high perishability. The goal is to demonstrate scalable strategies for better demand response, reduced waste, and optimized pricing and revenue.

In a nutshell, three core challenges have been highlighted in this literature review for perishable goods retail: volatile demand, constrained shelf life, and the inherent complexity of pricing and supply chain decisions. However, a significant gap still remains. Despite advances in forecasting, inventory, and pricing models, but research still lacks a unified, AI-driven framework that designed especially for perishable FMCG, particularly dairy items. The purpose of this project is to address these gaps with a practical, data-driven solution. By integrating granular sales data with predictive analytics and AI support, we aim to improve efficiency, reduce waste, and optimize revenue for perishable dairy goods in a retail context. This responds to known challenges and offers dual value to theory and practice.

3. Problem Statement

3.1 Challenges in Managing Perishable Dairy Goods

Short shelf life, high demand variability, and tight profit margins present an inherent operational challenge in managing the perishable dairy products. Given these challenges, the

demand forecasting, replenishment, pricing and waste reduction must be coordinated by all retailers, whereas traditional decision-making methods often address these factors independently. This ultimately leads to frequent understocking, overstocking, and spoilage. Research confirms that demand for perishable goods is both highly seasonal and volatile. Conventional forecasting techniques often fail to accurately capture these rapid market fluctuations (Zhu, 2024; Ye et al., 2024).

3.2 Limitations in Inventory and Expiry Management

Besides that, managing inventory and expiration dates of dairy products is also a key operational challenge. It becomes more difficult by manual practices and static safety stock methods. Without a real-time spoilage monitoring, they lead to avoidable waste (Birkmaier et al., 2024; Falayi et al., 2024). Although tools like predictive models and real-time data integration have been shown to improve performance and stock optimization (Duttyal, 2025), however, many retailers do not have the necessary analytical infrastructure to implement them (Korie, 2025; Falayi et al., 2024).

3.3 Gaps in Pricing Strategies for Perishable Goods

The literature presents a clear but unresolved challenge. Proven benefits of dynamic pricing for revenue and freshness (Korie, 2025; Zhu, 2024) are undermined by its typical isolation from essential forecasting and inventory management frameworks, a systemic gap identified by Komijani and Sajadieh (2024) and reiterated by Korie (2025).

3.4 Limited Adoption of AI and Predictive Analytics

AI holds strong potential to solve interconnected retail challenges, such as forecasting, spoilage prediction, inventory optimization, and dynamic pricing. However, adoption remains limited. Implementation is hindered by challenges like data scarcity, model interpretability issues, and system integration complexities, as noted by Korie (2025), Falayi et al. (2024), and Birkmaier et al. (2024).

3.5 Research Gap and Project Focus

In response to these limitations, this study pursues a clear objective, which is to develop an integrated, AI-driven decision-support framework for perishable dairy. The purpose is to apply AI analytics to a Dairy Product Sales dataset to directly enhance demand forecasting, expiry management, and pricing. Through predictive and prescriptive modeling, the work seeks to

reduce waste, optimize inventory, and improve revenue, thereby advancing sustainable operations.

4. Data Preparation

To prepare the data for the modelling process, several cleaning steps were carried out beforehand. The Date, Production Date, and Expiration Date fields were reformatted into a consistent datetime structure for use in later steps. Dates that were difficult to read or interpret were corrected or replaced to prevent errors during data processing.

To support the classification task, a new feature named LowStock was created. The LowStock feature assigns a value of 1 to items with a Quantity in Stock below the Minimum Stock Threshold. Items that do not meet this condition are assigned 0. This feature made it easier to identify products that might need replenishment.

Key numerical features such as production volume, shelf life, revenue, and stock levels were included in the dataset. Categorical features including product name, location, brand, and storage condition were also incorporated. These features were processed with a ColumnTransformer. Numerical values were scaled with StandardScaler to place them on a similar scale. The categorical fields could not be used directly in the model. They were processed with OneHotEncoder, which produced several binary columns for each category. These steps produced a more consistent dataset for training.

The dataset was split into training and testing sets using an 80:20 ratio. This was the final step in the preparation process. The LowStock target was stratified so that both sets had similar proportions of low-stock and regular-stock items. These steps prepared the dataset for further analysis and model development.

5. Model Building, Training, and Testing

Three machine learning models were created as part of the analysis of the dairy inventory dataset. Each model was trained and tested on the cleaned data, with each one focusing on a different aspect of the dataset.

5.1 Demand Forecasting Using a Seven-Day Moving Average

The daily demand values were chronologically arranged, and a seven-day moving average was computed to highlight short-term fluctuations. To avoid data leakage, the moving average was shifted forward by one day. The time series was split into an 80:20 ratio for training and testing. This method provided a smoothed signal that improved the stability and reliability of the demand forecast.

5.2 Low Stock Classification Using Logistic Regression

The Logistic Regression model was trained on a combination of numerical and categorical variables to classify items at risk of falling below minimum stock thresholds. To maintain the original class distribution, stratified sampling was used when generating the training and testing sets. Model performance was assessed using accuracy, precision, recall, and the F1-score to describe how well the model detected low-stock items.

5.3 Pricing Prediction Using Linear Regression

A Linear Regression model was created to model the Price per Unit using operational and product-related features. The model was trained on a portion of the dataset and tested on the remaining data. Model performance was reported using MAE, RMSE, and R-squared to describe the closeness of the predictions to the actual prices and the proportion of price variation captured by the model.

6. Graphs and Model Metrics

6.1 Demand Forecasting Graph

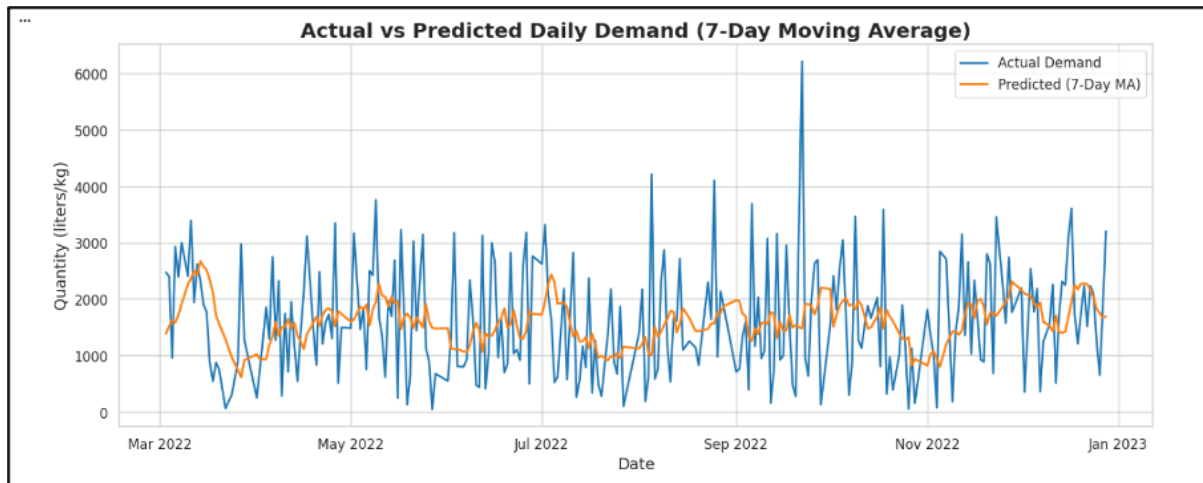


Figure 1. Actual and predicted daily demand using a seven-day moving average.

The graph compares actual demand with values estimated using a seven-day moving average. Similar movements in both lines indicate that the model reflects the overall trend. MAE and RMSE provide an indication of how well the forecasts align with the real demand.

6.2 Low-Stock Classification Model Performance

Confusion Matrix

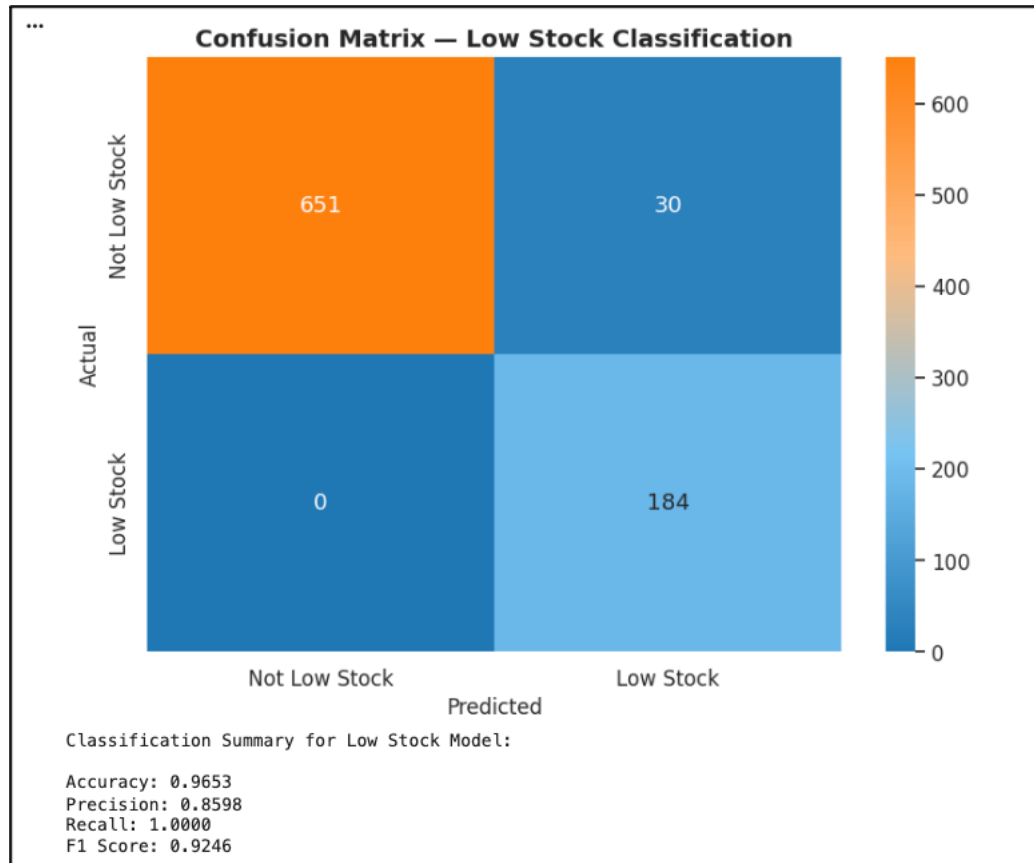


Figure 2. Confusion matrix for the low-stock classification model.

The confusion matrix tells us which predictions were correct and which ones weren't. Because the model produces so few false negatives, it rarely fails to identify low-stock items. This is important for making restocking decisions on time.

ROC Curve

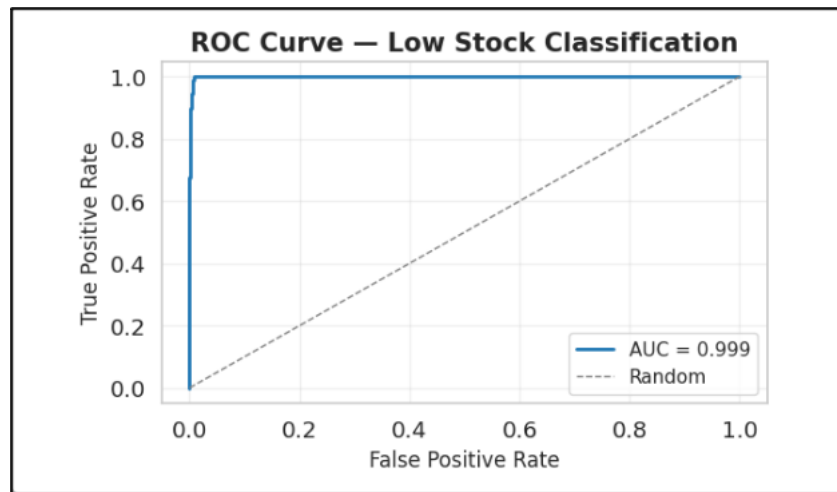


Figure 3. ROC curve and AUC performance of the low-stock classification model.

The ROC curve shows how well the model distinguishes low-stock items from those with sufficient stock. Since the AUC is high and close to 1, the model appears to be effective at separating the two groups.

Feature Importance Table

	Feature	Coefficient	Importance ($ \beta $)
1	Quantity in Stock (liters/kg)	-12.648682	12.648682
2	Quantity (liters/kg)	-6.282933	6.282933
3	Quantity Sold (liters/kg)	4.569076	4.569076
4	Minimum Stock Threshold (liters/kg)	2.074622	2.074622
5	Product Name_Butter	-0.525412	0.525412
6	Customer Location_Rajasthan	0.525379	0.525379
7	Brand_Passion Cheese	0.456284	0.456284
8	Price per Unit	-0.437554	0.437554
9	Customer Location_Karnataka	-0.426786	0.426786
10	Brand_Britannia Industries	-0.426471	0.426471
11	Customer Location_Tamil Nadu	-0.420500	0.420500
12	Customer Location_Haryana	0.416623	0.416623
13	Brand_Parag Milk Foods	0.390321	0.390321
14	Price per Unit (sold)	0.341089	0.341089
15	Product Name_Ice Cream	-0.304332	0.304332

Figure 4. Ranked feature importance table for the low-stock classification model.

This table shows which features have the strongest role in predicting low-stock items. Features with larger absolute coefficient values have a stronger influence on the prediction outcome.

Feature Importance Visualization

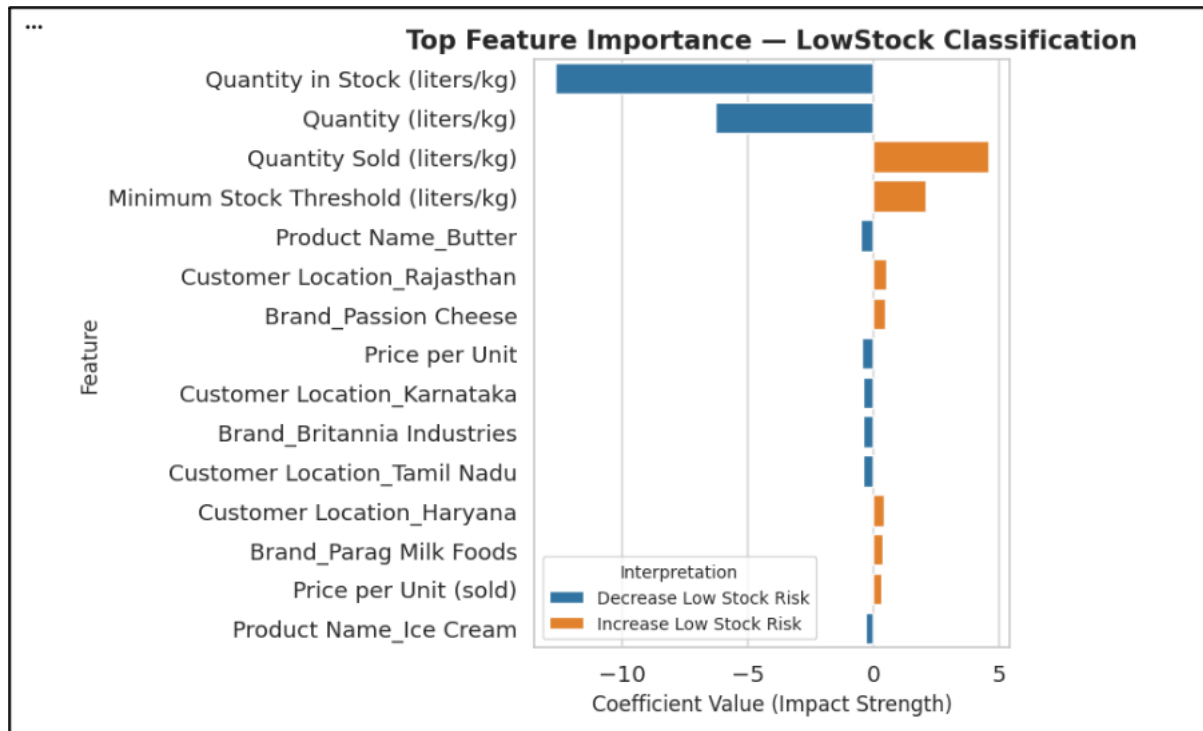


Figure 5. Top feature impact plot for the low-stock classification model.

The bar chart shows which variables increase or decrease the likelihood of items being classified as low stock. Orange features indicate an increased likelihood of being classified as low stock, while blue features indicate a reduced likelihood. The bar size reflects the strength of feature impact.

6.3 Pricing model results

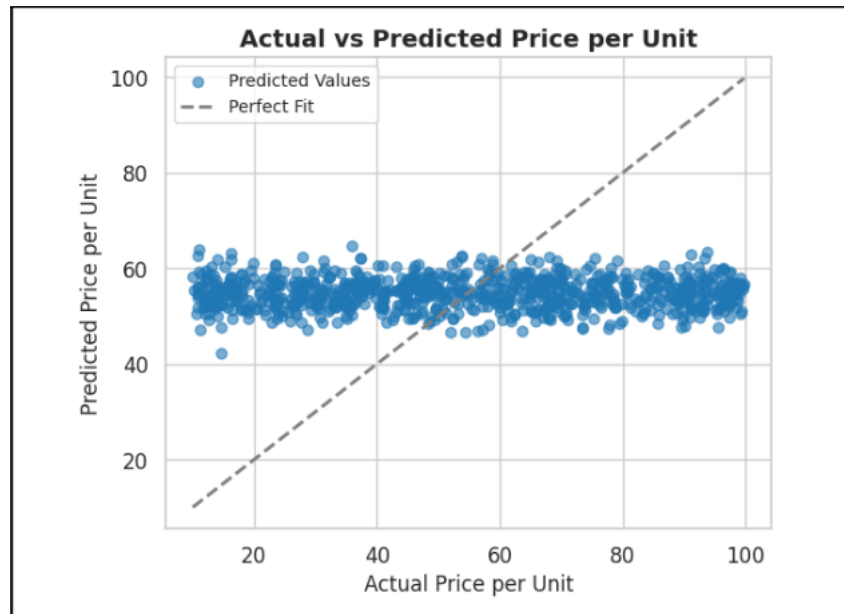


Figure 6. Actual vs. predicted unit prices using the linear regression model.

This scatterplot compares the model's predicted prices with the actual prices. Instead of forming a tight line along the diagonal, the points are spread out, showing that the predictions are not very close to the real values. The evaluation metrics (MAE, RMSE, and R-squared) support this by showing that the model only explains a small part of how prices change. This means the model may need more features or a different approach to improve accuracy.

7.Implementation

Python was used to develop and run the models on Google Collab. Pandas and NumPy handled data preparation, while Matplotlib and Seaborn were used to generate the visualisations. The machine learning models were built using Scikit-learn, where numerical and categorical features were processed using StandardScaler and OneHotEncoder. The same preprocessing was applied to both the training and testing data to avoid data leakage. Once the models were trained, their results were checked. Precision, recall and the F1-score were used to evaluate the classification model, while MAE, RMSE and R-squared were used for the forecasting and pricing models.

8. Solution Output Using Graphs

The graphs generated by the models illustrated how the system can assist with daily decisions in dairy inventory management. The demand forecasts showed when production should be increased or slowed down. This reduces the risk of producing too much or failing to meet daily requirements. The low stock predictions identified items that were running low. This allows staff to restock them in time and prevent shortages. The pricing prediction model determines suitable prices by considering production factors, product characteristics, and stock levels.

Overall, the graphs made the results easier to understand and showed how the models can help manage inventory more smoothly and reduce losses.

The graphs used in this section are shown in Figures 1 to 6.

9. Model Deployment Using Gradio

For deployment, the models were integrated into an interactive interface using Gradio so that predictions can be generated without writing code. The interface includes three modules, each linked to one of the machine learning models developed earlier.

The Demand Forecast module requires fewer user inputs compared to the other two models because it follows a time-series forecasting approach. This model relies on past sales records to generate the forecast. Product characteristics and farm information are not used in this process. Instead of using relationships between multiple data fields, the model generates its prediction from changes in sales over time, and the user's only role is to select the relevant historical records.

Table 1. Overview of the deployed Gradio modules and their corresponding machine learning models.

Gradio Tab	Model Type	Purpose
Low Stock Prediction	Classification (Logistic Regression)	Identifies whether an item is at risk of falling below the minimum stock level.

Price Estimator	Regression (Linear Regression)	Estimates the expected selling price of a dairy product based on production and product attributes.
Demand Forecast	Time-Series Moving Average Forecasting	Predicts expected product demand over future days based on historical patterns.

Each tab allows users to enter the required inputs through fields such as numerical boxes and dropdown menus. After the values are submitted, the interface displays the model output together with the corresponding visualisation. This design enables the system to be used by operational staff as well as technical users. The code used to initialise and deploy the final Gradio application in Google Colab is shown below:

```
# Reset any previous Gradio sessions
gr.close_all()
```

Once executed, Gradio generates a public link that can be accessed from any connected device. This demonstrates how the deployed solution can be used in practical settings where timely decisions are needed. Screenshots of the deployed interface and the three tabs are presented in Figures 7–10.

[illegible][illegible]

Figure 8. Price Estimator tab interface showing operational input fields and the resulting price prediction output.



Figure 9. Demand Forecast tab interface showing historical input selections and the resulting time-series forecast and visual output.

The deployment is complete, and the system is now available for use.

10. Conclusion

This study has successfully addressed the critical operational challenges inherent in the dairy retail sector, specifically the volatility of demand, the constraints of short shelf life, and the complexities of pricing strategy through the development of an integrated, data-driven framework. By leveraging the Dairy Product Sales dataset (Suraj, 2023), this project moved beyond theoretical discussion to provide empirical evidence on how Artificial Intelligence (AI) can resolve the supply-demand mismatches that characterize the Fast-Moving Consumer Goods (FMCG) industry.

The analytical results validate the efficacy of the proposed machine learning models in replacing rigid, manual inventory practices. The seven-day moving average model demonstrated that smoothing short-term fluctuations provides a reliable baseline for demand forecasting, directly addressing the unpredictability issues raised by Zhu (2024). Furthermore, the application of Logistic Regression for low-stock classification proved highly effective, yielding a high Area Under the Curve (AUC) and minimal false negatives. This specific capability answers the call by Birkmaier et al. (2024) and Falayi et al. (2024) for proactive intervention systems that can prevent stockouts before they occur. Additionally, the Linear Regression model for pricing established a robust link between product attributes and value, offering a systematic solution to the revenue optimization gaps identified by Korie (2025).

The significance of this research lies in its integrated approach. While previous literature often treated demand forecasting, inventory management, and pricing as isolated functions, this study demonstrates that they are deeply interconnected variables within the supply chain. By utilizing a unified pre-processing pipeline, including feature engineering for low-stock alerts and standardized scaling, the framework allows for a holistic view of retail operations. This directly bridges the research gap highlighted by Komijani and Sajadieh (2024) regarding the lack of comprehensive decision-making frameworks designed specifically for goods with stochastic lifespans.

In conclusion, the application of these models demonstrates that digital transformation is not an optional upgrade for dairy retailers, but rather a strategic necessity. Retailers can concurrently accomplish two competing goals by switching from experience-driven to forecast-driven planning: tightening supply-chain responsiveness to cut waste and guaranteeing product availability to maintain customer satisfaction. As the industry faces increasing pressure regarding sustainability and inflation, the AI-driven methodologies explored in this assignment provide a scalable, reproducible foundation for building a more resilient and profitable dairy supply chain.

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13. Code and analysis are available in the [Google Collab Notebook](#)